

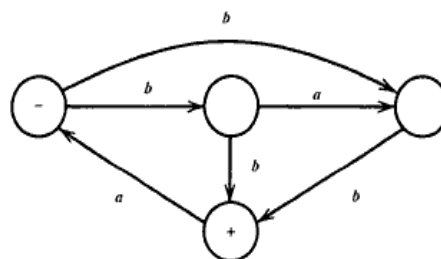
Unique Paper Code : 32341502
 Name of the Course : B.Sc. (H) Computer Science
 Name of the Paper : Theory of Computation
 Semester : V
 Year of admission : 2019 and onwards

Duration: Three Hours

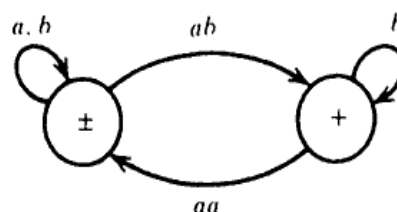
Maximum Marks: 75

Instructions for Candidates:

- i. Attempt any **FOUR** questions.
 - ii. Each question carries equal marks.
 - iii. Consider $\Sigma = \{a, b\}$ for all the questions unless specified otherwise.
1. Consider the language L, of all the words of length four or more having first two letters same as last two letters.
For the above language, perform the following:
 - Write all the words of L with the length five or less
 - Write the number of words having length six
 - Construct the regular expression
 - Build Finite Automaton (FA)
 2. Prove that it is true for all the regular languages that complement of a regular language is also regular.
Construct the deterministic finite automaton (DFA) that recognizes the same language as the non-deterministic finite automaton (NFA) given below and also describe the language recognized by it.



Convert the following transition graph into its equivalent regular expression:



3. Consider the following languages:

L_1 = Language of all the words having 'b' at second position

L_2 = Language of all the words having no two consecutive 'a's

Construct Finite Automaton FA_1 for L_1 , FA_2 for L_2 . Also construct regular expression and finite automata for the following:

- $L_1 + L_2$
- $L_1 \cap L_2$
- $(L_1)^*$

4. For the language $L_3: a^{n+m}b^m c^n$; where $\Sigma = \{a, b, c\}$ and $m, n \geq 1$, using pumping lemma, prove that the language is not regular. For the above language, do the following:

- Write a context free grammar (CFG) for L_3 , and construct parse tree for the word $aaabbc$ using this CFG
- Build a pushdown automaton (PDA) for L_3

5. Consider the following context free grammars (CFGs):

$G_1: \begin{aligned} S &\rightarrow bS \mid aX \\ X &\rightarrow bS \mid aY \\ Y &\rightarrow aY \mid bY \mid a \mid b \end{aligned}$

$G_2: \begin{aligned} S &\rightarrow XaX \mid bX \\ X &\rightarrow XaX \mid XbX \mid \Lambda \end{aligned}$

$G_3: \begin{aligned} S &\rightarrow A \mid AA \\ A &\rightarrow B \mid BB \\ B &\rightarrow abB \mid b \mid bb \end{aligned}$

$G_4: \begin{aligned} S &\rightarrow BABABA \\ A &\rightarrow a \mid \Lambda \\ B &\rightarrow b \mid \Lambda \end{aligned}$

For the above CFGs, perform the following:

- Write a regular expression for the language represented by G_1
- Convert G_2 into its equivalent CFG without null(Λ)-production
- Convert G_3 into its equivalent CFG without unit-production
- Convert G_4 into its equivalent Chomsky Normal Form (CNF)

6. Consider the language $L_4: a^n b^n c^n$ where $\Sigma = \{a, b, c\}$ and $n \geq 1$, and perform the following:

- Build a turing machine M_1 , that accepts L_4
- Build another turing machine M_2 , that accepts complement of L_4
- Is L_4 a recursive language or recursively enumerable language? Justify your answer
- Is L_4 a context-free language? Justify your answer.